THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The total number of triangles removed from the Sierpinski triangle up to stage 5 is: 1

(a) 243 **(b)** 242 **(c)** 121 **(d)** 81

2. A prism has 30 edges. The number of its faces is: 1

(a) 13 **(b)** 11 **(c)** 12 **(d)** 10

3. A pyramid has 13 vertices. The number of its edges is: 1

(a) 26 **(b)** 12 **(c)** 14 **(d)** 24

4. All 16 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 48 **(b)** 16 **(c)** 24 **(d)** 32

5. A polyhedron has 20 faces and 30 edges. Its number of vertices is: 1

(a) 12 **(b)** 10 **(c)** 52 **(d)** 48

6. A cube of side 9 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

(a) 721 **(b)** 343 **(c)** 294 **(d)** 49

7. A Koch curve starts with a line of length 36 cm. The total length after stage 2 (in cm) is: 1

(a) 72 **(b)** 20.25 **(c)** 96 **(d)** 64

8. The total number of triangles removed from the Sierpinski triangle up to stage 6 is: 1

(a) 243 **(b)** 728 **(c)** 729 **(d)** 364

9. A prism has 36 edges. The number of its faces is: 1

(a) 15 **(b)** 12 **(c)** 14 **(d)** 13

10. A pyramid has 16 vertices. The number of its edges is: 1

(a) 17 **(b)** 30 **(c)** 32 **(d)** 15

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A Koch curve starts with a line of length 81 cm. Find the number of segments and the total length at stage 2. 2

12. A cuboid is 20 cm long, 12 cm wide and 9 cm high. Find the sum of lengths of all its edges and the number of edges. 2

13. A prism has 48 edges. Find its number of faces and vertices. 2

14. A Koch curve starts with a line of length 90 cm. Find the number of segments and the total length at stage 2. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 9 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3
(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

16. All 28 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

17. A Sierpinski triangle starts with a triangle of area 2048 cm². Find the shaded area and the number of shaded triangles at stage 5. 3

OR

A cube of side 10 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3
(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

18. All 30 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

OR

A Sierpinski triangle starts with a triangle of area 4096 cm². Find the shaded area and the number of shaded triangles at stage 3. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A prism and a pyramid both have a nonagonal base. Find the difference between the number of 5
(i) faces
(ii) vertices
(iii) edges of the prism and the pyramid.

OR

A Koch snowflake is built from an equilateral triangle of side 45 cm. Find its perimeter at stage 0, stage 1 and stage 2. 5

20. A prism and a pyramid both have a decagonal base. Find the difference between the number of 5
(i) faces
(ii) vertices
(iii) edges of the prism and the pyramid.

OR

A Koch snowflake is built from an equilateral triangle of side 54 cm. Find its perimeter at stage 0, stage 1 and stage 2. 5

END OF QUESTION PAPER **Total: 40 Marks**